

HOMOLOGICAL LIOUVILLE PATH ENSEMBLES AND PROJECTIVE EMPIRICAL-PROCESS LARGE DEVIATIONS

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ABSTRACT. For a compact finite-horizon periodic Lorentz family we construct a renormalized physical trace bundle through branch births, compute a finite family of explicit periodic arithmetic certificates, and prove geometric temporal nonintegrability from returned inverse branches. A four-range Fourier theorem, including a separate very-high-frequency integration-by-parts regime, yields the vector-root density local limit theorem and correctly separated local, central, saturated, and macroscopic window asymptotics.

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1. ROUND-TEN RECONSTRUCTION: A VERIFIED BILLIARD PACKET AND A FOUR-RANGE FOURIER THEOREM

We use the triangular lattice with basis $e_1 = (1, 0)$, $e_2 = (1/2, \sqrt{3}/2)$, one circular scatterer of radius R , and

$$K = [9/20, 47/100].$$

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The finite-horizon collision map, its homological displacement $\kappa_R \in \mathbb{Z}^2$, and the roof τ_R are those of the paper. Every numerical inequality below is recorded in `A2_ROUND10_CERTIFICATE.json` and checked by `tools/verify_a2_round10_certificate.py`.

1.1. A birth bundle with no fictitious physical inverse. At a simple branch birth write $r = R - R_0$. On the birth side the two branch traces have the expansion

$$T_{r,\pm} = T_0 \pm r^{1/2}T_1 + rT_2 + O(r^{3/2}).$$

Introduce coefficient coordinates

$$U^{\text{ev}} = \frac{1}{2}(T_{r,+} + T_{r,-}), \quad U^{\text{od}} = \frac{1}{2r^{1/2}}(T_{r,+} - T_{r,-}) \quad (r > 0).$$

For $r \leq 0$, the coefficients are retained as virtual normal jets but their newborn physical realization is zero. The bundle norm is a coefficient norm, not the norm of an inverse of the physical realization.

Lemma 1.1 (Renormalized one-sided birth bundle). *There is a fixed coefficient space $\mathbb{B}_{\text{bl}}^j$ and bounded realization maps $E_R : \mathbb{B}_{\text{bl}}^j \rightarrow \mathcal{D}'$ such that:*

- (i) *on the birth side the two realized traces are $U^{\text{ev}} \pm r^{1/2}U^{\text{od}}$;*
- (ii) *on the no-branch side the newborn realization is zero;*
- (iii) *in $s = \text{sgn}(r)|r|^{1/2}$, the realization is C^j from each strong coefficient level to the next weak physical level; and*
- (iv) *the transfer operator preserves the closed graph $\{(U, E_R U)\}$ and satisfies a uniform graph Lasota–Yorke estimate.*

No uniformly bounded inverse from a vanishing physical current to an order-one coefficient is asserted.

Proof. The parameterized Morse lemma puts the tangency equation into fold form $\zeta^2 = s^2$. Taylor expansion in s , not in $r = s^2$, gives the even and odd jets with uniform remainder. Stable-curve trace estimates place the jet coefficients one strong order below the bulk and the remainder in the weak level. The branch transfer formula is smooth in the blown-up variables; on bulk plus jets it is a finite lower-triangular system. Uniform finite horizon, curvature, homogeneity complexity and one-step expansion give the graph Lasota–Yorke inequality, while stable-curve and strip truncation give compactness. The virtual no-branch coefficient is never physically inverted. $\square$

Theorem 1.2 (Uniform vector–roof spectral graph). *For a fixed finite jet order, the operators*

$$\mathcal{L}_{R,z,b}f = \mathcal{L}_R(e^{z \cdot \kappa_R + ib\tau_R} f)$$

act on the graph bundle of Lemma 1.1. On one common complex neighborhood of $(z, b) = (0, 0)$ they have a simple analytic leading eigenvalue and a uniform spectral remainder. Material derivatives in R lose one strong level per derivative. Fixed central cylinder insertions and their material derivatives act on the same scale.

Proof. The vector factor is branchwise constant and the roof factor has the stable oscillation and unstable holonomy bounds of the Jacobian. In fold coordinates the two newborn multipliers form an even/odd Taylor system and preserve the physical graph. The graph Lasota–Yorke estimate, compact strong-to-weak embedding and mixing yield a uniform Riesz contour. Analytic perturbation and the material resolvent identity give the derivatives. A central cylinder is cut by its finite singularity preimages, so all of its one-sided traces are among the declared jet coordinates. $\square$

1.2. An explicit arithmetic packet. There are two primitive winding one-collision orbits with homologies e_1, e_2 . Let w_0 be the two-collision nearest-neighbour backtracking orbit. Let w_3 circulate around one triangular gap and w_4 circulate around the rhombus with centers $0, e_1, e_1 + e_2, e_2$. Their roof sums and collision counts are

w	$\tau_R(w)$	$n(w)$
w_0	$2(1 - 2R)$	2
w_3	$3(1 - \sqrt{3}R)$	3
w_4	$4\sqrt{1 - (1 + \sqrt{3})R + 2R^2}$	4.

Lemma 1.3 (Verified four-coordinate periodic certificate). *The five orbits $w_0, \dots, w_4$ are regular and unobstructed for every $R \in K$. If $c_R(w) = (\kappa_R^{(1)}, \kappa_R^{(2)}, \tau_R, n)(w)$, then*

$$\left| \det(c_R(w_j) - c_R(w_0))_{j=1}^4 \right| = |2\tau_R(w_3) - \tau_R(w_4) - \tau_R(w_0)| \geq 0.55.$$

Consequently the joint lattice/roof cocycle has no nontrivial compact-frequency character.

Proof. For the triangular orbit the contact points lie on the inward centroid bisectors and every leg has length $1 - \sqrt{3}R$. For the rhombus the four inward angle-bisector contacts give the displayed common leg length. Direct substitution verifies equal incident and reflected angles. Distance-to-segment minimization against the finite neighboring lattice list gives clearances at least 0.16 and 0.058, respectively. The first two rows of the four-by-four difference matrix form the homology identity. The remaining scalar determinant is evaluated on subintervals of width 10^{-4} , with a mean-value derivative enclosure; the certified absolute lower bound is 0.55. A compact character coboundary vanishes on every periodic sum and is therefore zero. $\square$

1.3. A geometric UNI pair independent of arithmetic. The clockwise and counterclockwise three-collision branches around the same triangular gap return to one common collision rectangle.

Theorem 1.4 (Verified returned-branch UNI). *There are a fixed unstable interval J , a return time N , and inverse branches $h_R^\pm : J \rightarrow J$, continuous in R , such that*

$$\inf_{R \in K} \inf_{x \in J} \left| \partial_x (\tau_N(h_R^+ x) - \tau_N(h_R^- x)) \right| \geq 3/4.$$

The branch words, interval endpoints, cone bounds and derivative intervals are part of the certificate.

Proof. The billiard generating function is the flight length, and its derivative in the initial arclength is the tangential momentum. At the symmetric triangular orbit the clockwise and counterclockwise branches have momenta $\pm 1/2$, so the temporal-distance derivative is one. The explicit flight Hessians on K bound its variation by $1/8$ on the certified interval. The common return suffix changes it by at most $1/8$, by inverse-branch cone contraction. The clearance bounds keep both branches away from grazing and all other singularities. $\square$

1.4. Moderate and very-high frequency estimates.

Theorem 1.5 (Four-range twisted estimate). *For every integer $M \geq 6$ there are $C, c, A, \delta > 0$, uniform in $R \in K$, such that:*

- (i) *for $|(u, b)| \leq \delta$, the simple eigenvalue has the uniform cubic Gaussian expansion;*
- (ii) *on compact nonzero frequencies the spectral radius is at most $1 - c$;*
- (iii) *for $2 \leq |b| \leq e^{\delta n}$,*

$$\|\mathcal{L}_{R,iu,ib}^n\|_{\text{strong} \rightarrow \text{weak}} \leq C|b|^A e^{-cn/\log(2+|b|)},$$

- (iv) *for $|b| > e^{\delta n}$, after M roof integrations by parts,*

$$\|\mathcal{L}_{R,iu,ib}^n\|_{\text{insertion}} \leq C_M e^{C_M n} |b|^{-M}.$$

The last estimate also holds with a fixed central cylinder insertion.

Proof. The first two assertions follow from Theorem 1.2 and Lemma 1.3. The UNI pair, standard-family return and bounded distortion yield Dolgopyat cancellation in the moderate range. In the very-high range decompose into homogeneous inverse branches and integrate the oscillatory integral M times in the UNI coordinate. Derivatives of amplitudes and inverse roof derivatives are bounded by the finite jet scale; the number of branches grows at most exponentially in n . Since $|b| > e^{\delta n}$, choosing $M > C_M/\delta + 3$ makes the tail integrable. Thus the moderate estimate is never extrapolated to infinite frequency. $\square$

1.5. The corrected vector-roof local limit theorem. Let $K_n \in \mathbb{Z}^2$, $T_n \in \mathbb{R}$, let Σ_R be their joint covariance under a regular real tilt, $\Sigma_{K,R}$ its lattice block, and $\sigma_{c,R}^2$ the conditional roof variance.

Theorem 1.6 (Uniform density LLT and separated window regimes). *Uniformly for $|k_n - n\bar{\kappa}_R| = O(\sqrt{n})$:*

- (a) *on every fixed compact u -set,*

$$p_{n,R}(k_n, n\bar{\tau}_R + \sqrt{n}u) = \frac{e^{-\delta k_n^T \Sigma_{K,R}^{-1} \delta k_n / (2n)}}{(2\pi n)^{3/2} \sqrt{\det \Sigma_R}} e^{-u^2 / (2\sigma_{c,R}^2)} (1 + O(n^{-\eta}));$$

- (b) if $\Delta_n = o(\sqrt{n})$, a roof-sum interval of half-width Δ_n has the two-dimensional lattice prefactor n^{-1} multiplied by $2\Delta_n/(\sqrt{2\pi n}\sigma_{c,R})$;
- (c) if $\Delta_n/\sqrt{n} \rightarrow \lambda \in (0, \infty)$, the lattice prefactor is multiplied by the corresponding conditional Gaussian interval mass;
- (d) if $\Delta_n/\sqrt{n} \rightarrow \infty$ and $\Delta_n = o(n)$, that mass tends to one;
- (e) a fixed interval in the roof average is governed by a saddle-varying Bahadur–Rao integral and is not obtained by integrating a central Gaussian outside its uniform range.

All statements remain valid with a fixed central cylinder insertion.

Proof. Fourier inversion is split into the four ranges of Theorem 1.5. The small range gives the joint Gaussian density; the compact and moderate ranges are negligible; the very-high range is absolutely integrable after M integrations by parts. Integrating over subcentral roof intervals gives (b)–(d), with the n^{-1} lattice coefficient separated from the conditional roof probability. For a fixed average interval, cover the range by real source charts, tilt at the local saddle, and apply Laplace’s method to the local density theorem. The insertion statement follows from Theorem 1.2. $\square$

SCOPE AND VERIFICATION STATUS

This manuscript states positive theorem closure in the declared model-specific regular regime. Cross-paper inputs are used only in the acyclic order recorded in the round-ten dependency ledger. Repository source, proof-structure, counterexample, and build gates are internal reproducibility checks; they do not replace independent external mathematical review.

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